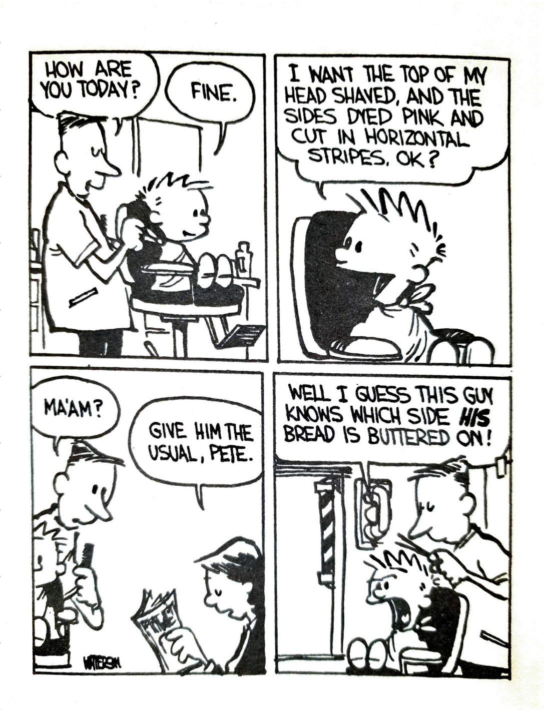
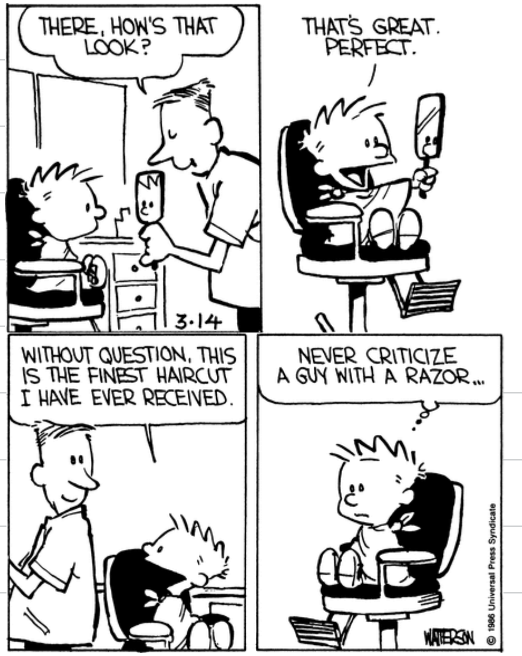
Recurrent Neural Networks - RNN

May 19, 2026

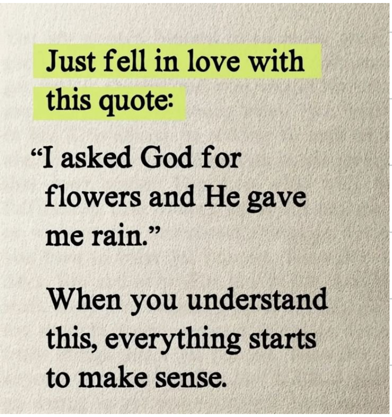
We will start 8:40 PM





Agenda

1. RNN Architecture
2. Training a Many to One RNN model
3. Expanding training methods to other types of RNN
4. Transformers Intro

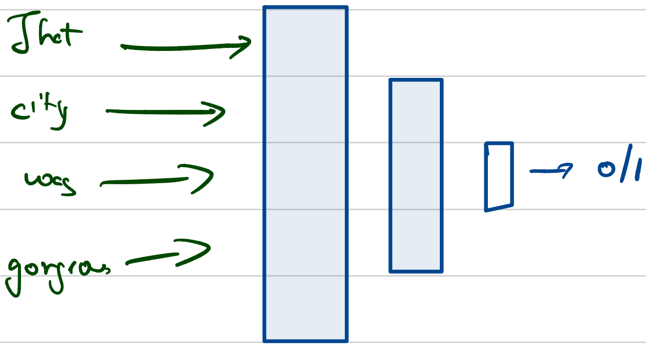


Just fell in love with this quote:

"I asked God for flowers and He gave me rain."

When you understand this, everything starts to make sense.

That city was gorgeous. — 0/1



Problems we're trying to solve with RNN:

1. Sequence isn't effectively captured.
2. My model gives equal weightage to all the words.
3. Negation
4. There is no connection between words.

Recurrent Neural Network

PBM – $\frac{12}{6+1}$

serving output

Previous time step

output

Current input

t=0

ghee/butter + Fumes

NULL

ghee/butter

Butter + Ghee

t=1

Roasted Spices

Spices

Hing + Jeera - Prefusi

t=2

More Roasted Spi

Spic

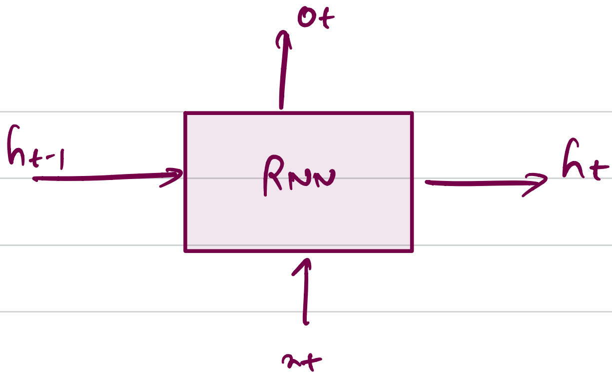
Khada masal

Pridu

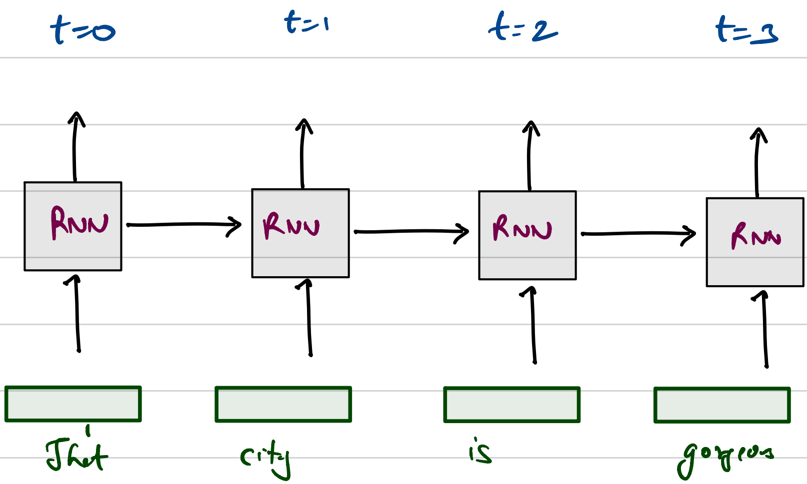
t=4

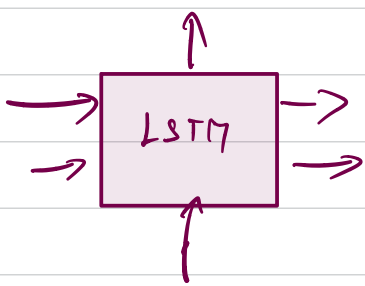
Cooked Puree

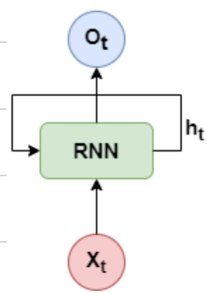
Puree - Snake

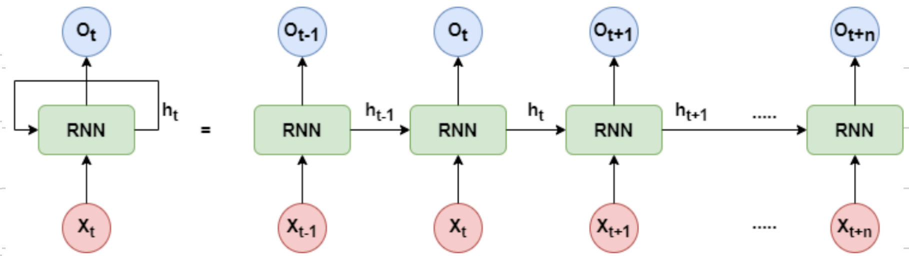


That city is gorgeous.





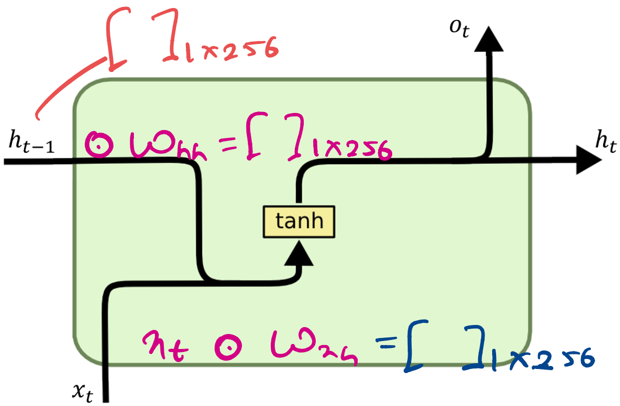




⊙ matrix multiplication

hidden dim: 256

embedding dim: 100



$$W_{xh} = [\ ]_{100\times256}$$

$$W_{hh} = [\ ]_{256\times256}$$

$$W_{ot} = [\ ]_{256\times256}$$

$[\ ]_{1\times100}$

$$O_t = \underbrace{h_t}_{[\ ]_{1\times256}} \odot \underbrace{W_{ot}}_{[\ ]_{256\times256}} = [\ ]_{1\times256}$$

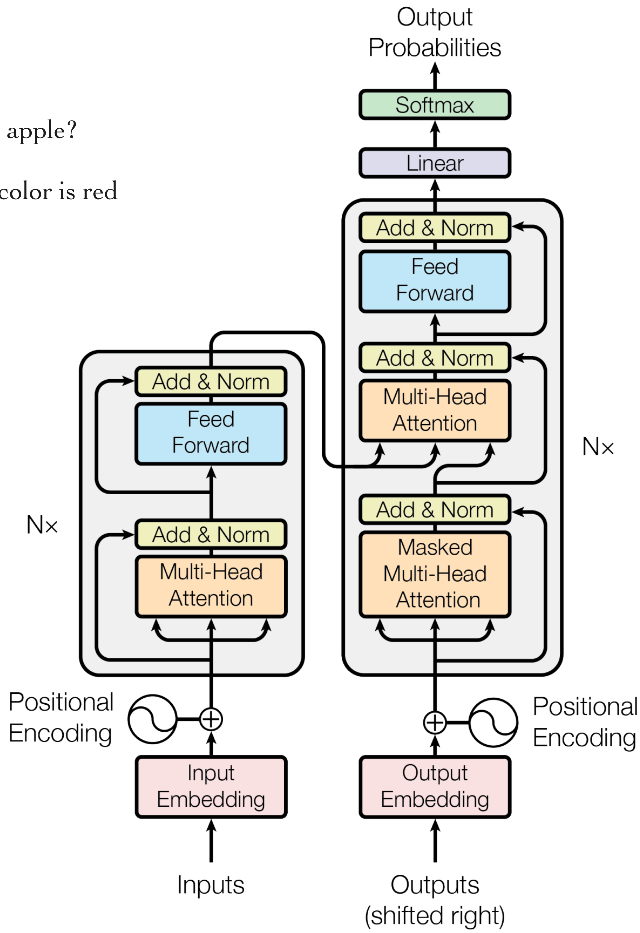
$$\tanh\big(\underbrace{x_t \odot W_{xh}}_{[\ ]_{1\times256}} + \underbrace{h_{t-1} \odot W_{hh}}_{[\ ]_{1\times256}} + \underbrace{b_y}_{[\ ]_{1\times256}}\big) = [\ ]_{1\times256}$$

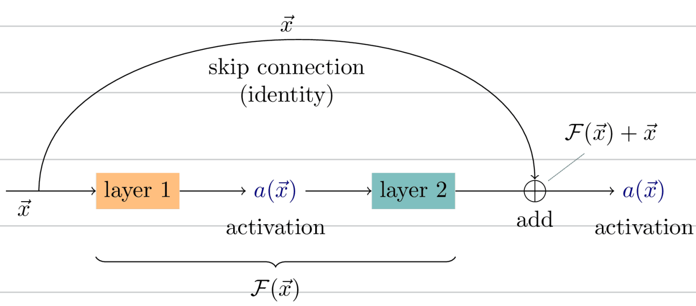
$$h_{t-1}||x_t = 1\times356 \odot W[\ ]_{356\times256}$$

$$= 1\times256$$

What color is apple?

Output: The color is red





c

Trf

What is appx? <ANS> Apple

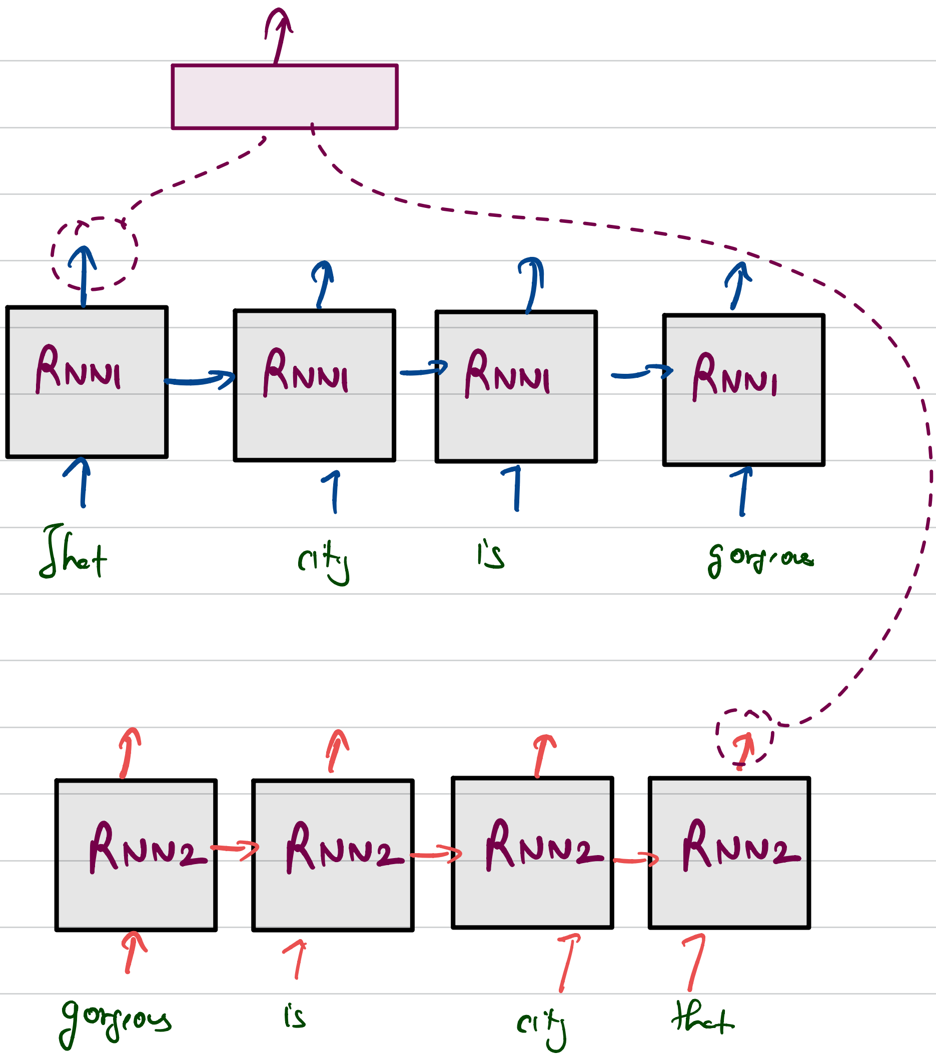
<EOS>

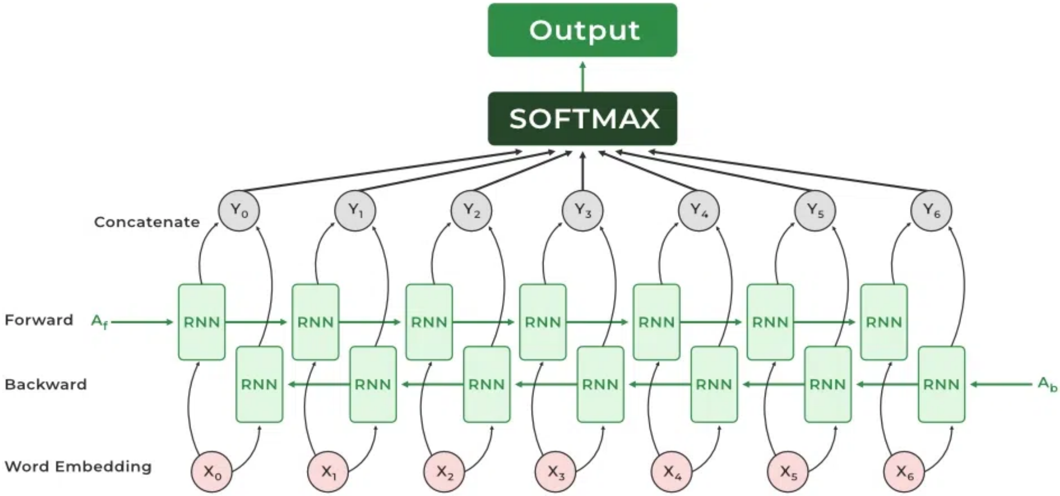
t=1

What is appx? <ANS> Apple

Bi-Directional RNN

I love visiting banks, as I love seeing fishes in clear blue water.





	f_1	f_2	f_3	f_4	f_5	f_6	y
x_1							
x_2							
x_3							
x_4							
x_5							
x_6							
x_7							
x_8							

batch size : b

Featur : m

batch_x : $b \times m$

batch_y : b

["I" , "love" , "cats"]

embedding dim : 128

3×128

batch of data - $b \times 3 \times 128$

batch_size x no_of_words x embedding_dim

1. A registration fee of 3540/-
2. It is scheduled for **26th to 28th June 2026**.

The tentative schedule is as follows:

- **26th June 2026** – Check-in & accommodation allotment
- **27th June 2026** – Campus Immersion Day 1
- **28th June 2026** – Campus Immersion Day 2